12.4 Gastrointestinal Tract Infections

Human systems function by the energy produced from the digested food molecules. The food is swallowed through mouth and digested in the gastro intestinal tract. The food we consumed should be free of contaminations. The contaminated food causes gastrointestinal infections.

Through contaminated food and water the pathogens are ingested and they enter the GIT. In the small intestine they initiate an infection. Many times the pathogens that cause intestinal infections multiply in the GIT and produce their pathogenic effect in the intestine itself. Example: Shigellosis, Cholera.

The gastrointestinal tract (GIT) or alimentary canal includes the mouth, pharynx, throat, oesophagus (food Tube lead to the stomach), stomach, small and large intestine. It also includes accessory structures salivary glands, liver, gall bladder and pancreas lying outside the GIT. Secretions of these organs enhance the digestion of food molecules (Figure 12.11).

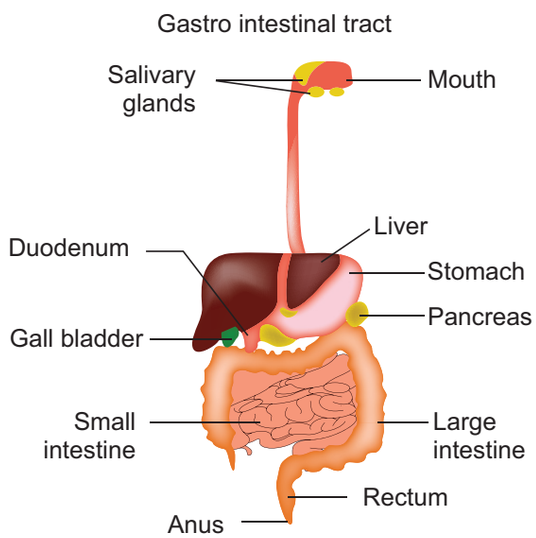


Figure 12.11: The structure of Gastro intestinal tract

Difference between infection and intoxication

Microbial diseases of digestive system are typically transmitted by a fecal oral route. Most such diseases result from the ingestion of food or water contaminated with pathogenic microorganisms or their toxins. These pathogens usually enter the food or water supply after being shed in the feces of people or animals infected with them.



- Each milliliter of saliva can contain millions of bacteria
- Stomach/small intestine has very few microorganisms because of hydrochloric acid present in the stomach.
- Large intestine harbours microbial population exceeding 100 billion of bacteria per gram of feces (40% fecal masses contain microbial cell material)
- Large intestine microbial population mainly contain anaerobes and facultative anaerobes.

After ingestion of pathogenic microorganisms, localization and multiplication of organisms takes place in the GIT and is called infection. Microorganisms may penetrate into intestinal mucosa and grow there or they may penetrate to other organs. Gastroenteritis is usually classified as either infection or intoxication. Food borne diseases can arise from either infection or intoxication. In both cases, bacterial toxins are typically responsible for producing disease signs and symptoms. In a food infection the microbial agent ingested

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